

081 Build your own barometer

45
MINS

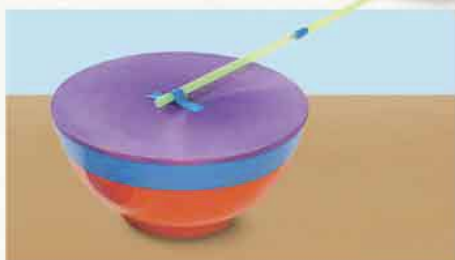
As air gets hotter or colder, it causes shifts in the air pressure, which impacts weather patterns. Build a barometer to measure how the pressure changes in the air—and you can also predict the weather.

What you need

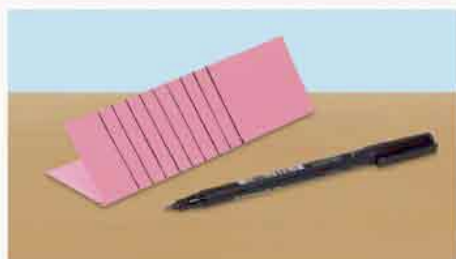
- Bowl • Large balloon
- Scissors • Tape
- 2 drink straws
- Colored posterboard
- Markers • Ruler



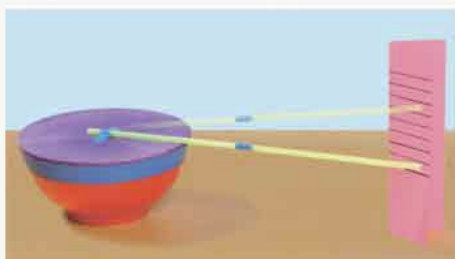
1 Cut off most of the round part of the balloon and stretch it tightly over the bowl. Attach it securely with tape.



2 Slit the end of one straw, push it inside the other straw, and tape together. Tape one end to the middle of the balloon.



3 Using your marker and ruler, draw a series of lines $\frac{1}{4}$ in (6 mm) apart on a folded piece of posterboard. Stand the scale by the end of the pointer.



4 Check the scale every few hours to see if the pointer has moved with changes in the air pressure. Up indicates clear weather; down shows wet weather.

082 Measure hailstones

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Center a square block of 1 in (2.5 cm) thick foam on a large sheet of heavy foil, dull side up. Fold the foil over the edges and tape in place. Put the pad out when it hails and measure the dents.



Dull
side up

Large
hailstone
dent